

Number Series

Logical Reasoning & Aptitude Study Guide

Patterns, tricks and a 10-second exam strategy

Number Series is a common aptitude topic in the placement tests of TCS, Infosys, Cognizant, Accenture, Capgemini, Wipro, HCL and others. The main skill is to **identify the pattern between consecutive numbers quickly**.

What's inside

1. The golden checking order
2. Fourteen pattern tricks with worked examples
3. The 10-second exam checklist
4. What to memorize
5. Four-level practice plan

1. First Rule: Always Check in This Order

When you see a series like **2, 5, 8, 11, ?** don't randomly try patterns. Check systematically, in this order. It saves time in the exam.

#	Check	Looks like
1	Addition / subtraction	+5, +5, +5
2	Multiplication / division	$\times 2$, $\times 2$, $\times 2$
3	Increasing or decreasing differences	+3, +5, +7
4	Squares / cubes	4, 9, 16, 25
5	Prime numbers	3, 5, 7, 11
6	Alternate terms	two interleaved series
7	Mixed operations	$\times 2 + 1$, $\times 2 + 2$
8	Special patterns	Fibonacci, digit-based

2. The Tricks

Trick 1: Check the Difference

Subtract each number from the next.

Example: 7, 12, 17, 22, ?

- $12 - 7 = 5$, $17 - 12 = 5$, $22 - 17 = 5$
- So the next term is $22 + 5 = 27$

Answer: 27

Shortcut

If the numbers are changing slowly, check differences first.

Trick 2: Increasing Difference

Sometimes the difference itself changes.

Example: 2, 5, 10, 17, 26, ?

Numbers	Difference
2 to 5	+3
5 to 10	+5
10 to 17	+7
17 to 26	+9

Differences are 3, 5, 7, 9, so the next difference is 11. Therefore $26 + 11 = 37$.

Answer: 37

Trick

If the first difference isn't constant, write the second-level differences.

Trick 3: Multiplication

Example: 3, 6, 12, 24, ?

- $3 \times 2 = 6$, $6 \times 2 = 12$, $12 \times 2 = 24$
- Next: $24 \times 2 = 48$

Answer: 48

Shortcut

Look for $\times 2$, $\times 3$, $\times 4$, $\div 2$, $\div 3$, etc.

Trick 4: Multiplication + Addition

Very common in aptitude exams.

Example: 2, 5, 12, 27, 58, ?

- $2 \times 2 + 1 = 5$
- $5 \times 2 + 2 = 12$
- $12 \times 2 + 3 = 27$
- $27 \times 2 + 4 = 58$
- Next: $58 \times 2 + 5 = 121$

Answer: 121

Shortcut

Pattern is $\times 2 + 1$, $\times 2 + 2$, $\times 2 + 3$, $\times 2 + 4$... (the added number grows by 1).

Trick 5: Squares

Memorize: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100.

Example: 4, 9, 16, 25, 36, ?

- $2^2 = 4$, $3^2 = 9$, $4^2 = 16$, $5^2 = 25$, $6^2 = 36$
- Next: $7^2 = 49$

Answer: 49

Trick 6: Cubes

Know at least the first ten cubes:

Number	Cube
1	1
2	8
3	27
4	64
5	125
6	216
7	343
8	512
9	729
10	1000

Example: 8, 27, 64, 125, ?

These are $2^3, 3^3, 4^3, 5^3$. Next: $6^3 = 216$.

Answer: 216

Trick 7: Alternate Numbers

One of the most important tricks. Don't examine all numbers together.

Example: 2, 5, 4, 10, 8, 20, 16, ?

- Odd positions: 2, 4, 8, 16 (pattern $\times 2$)
- Even positions: 5, 10, 20, ? (pattern $\times 2$)
- So $20 \times 2 = 40$

Answer: 40

Shortcut

If the normal pattern isn't visible, try the 1st, 3rd, 5th, 7th terms and, separately, the 2nd, 4th, 6th, 8th terms.

Trick 8: Prime Numbers

Know the start of the sequence: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31 ...

Example: 3, 5, 7, 11, 13, ?

- Next prime after 13 is 17

Answer: 17

Trick 9: Difference of Squares

Some series use differences of +3, +5, +7, +9 ...

Example: 1, 4, 9, 16, 25, ?

- Differences: +3, +5, +7, +9, so the next is +11
- $25 + 11 = 36$

Answer: 36

Shortcut

You don't need to memorize the squares if you spot the odd-number differences.

Trick 10: Alternating + and -

Example: 10, 15, 12, 17, 14, 19, ?

- Pattern: +5, -3, +5, -3, +5, -3
- Next step is -3: $19 - 3 = 16$

Answer: 16

Trick 11: Alternating Multiplication

Example: 2, 6, 12, 36, 72, ?

- Pattern: $\times 3$, $\times 2$, $\times 3$, $\times 2$...
- Next step is $\times 3$: $72 \times 3 = 216$

Answer: 216

Trick 12: Fibonacci Pattern

Each number is the sum of the previous two.

Example: 1, 1, 2, 3, 5, 8, ?

- $1 + 1 = 2$, $1 + 2 = 3$, $2 + 3 = 5$, $3 + 5 = 8$
- Next: $5 + 8 = 13$

Answer: 13

Trick 13: Difference of Differences

Example: 5, 9, 17, 29, 45, ?

- First differences: +4, +8, +12, +16
- These increase by 4, so the next difference is +20
- $45 + 20 = 65$

Answer: 65

Trick 14: Digit-based Pattern

Sometimes the digits themselves matter.

Example: 12, 24, 48, 96, ?

- This one is simply $\times 2$ (next is 192).
- But another type might use the digit sum (12 gives $1 + 2 = 3$) or the digit product (12 gives $1 \times 2 = 2$).
- If ordinary operations don't work, inspect the individual digits.

Answer: 192 (for this series)

3. The Most Important Exam Strategy

Use this 10-second checklist on every number series question:

Step	What to check	Example
1	Check + / -	constant addition
2	Check differences	+2, +4, +6, +8 ...
3	Check $\times$ / $\div$	$\times 2$, $\times 3$, $\div 2$
4	Check $\times$ together with +/-	$\times 2 + 1$
5	Check squares / cubes	n^2 , n^3
6	Check alternate terms	1st, 3rd, 5th ... and 2nd, 4th, 6th ...
7	Check primes / Fibonacci	2, 3, 5, 7 / 1, 1, 2, 3, 5
8	Check digit patterns	digit sum or product

4. What You Should Memorize

Item	Memorize
Squares	1^2 to 20^2
Cubes	1^3 to 10^3
Prime numbers	At least 1 to 100
Powers of 2	2, 4, 8, 16, 32, 64, 128, 256, 512
Powers of 3	3, 9, 27, 81, 243, 729

These will significantly reduce your calculation time.

5. Practice Method: Learn in This Order

Level 1: Easy

- Constant addition
- Constant subtraction
- Constant multiplication
- Constant division

Level 2: Medium

- Increasing differences
- Decreasing differences
- Alternate patterns
- Squares
- Cubes
- Prime numbers

Level 3: Hard

- Multiplication + addition
- Multiplication - subtraction
- Alternating operations
- Second / third differences
- Mixed patterns
- Digit-based series

Level 4: Company-style

- Missing number series
- Wrong number series
- Multiple blanks
- Long series
- Time-based questions

Golden Rule

Don't try to guess the pattern immediately. **Calculate the differences first.** If that fails, check multiplication / division, then alternate terms.